

Sarah Benavides

CS 362 Lab 12

4/15/2026

Linux Forensics (Autopsy Platform)

Part 1: Non-Technical summary

This lab walked us step-by-step on how to use the Autopsy platform on a Linux system to perform digital forensic analysis in a simple yet structured way. We started this lab by installing and experimenting with Autopsy. We accessed the platform through a web browser, which provided a user-friendly interface for investigating digital evidence. Before continuing, a specific website was used to locate a digital image to download for the purpose of analyzing it within this platform. A new case was then created to organize the investigation, and a host system was added to store analysis results.

Once the image was loaded and the host information was entered, the tool allowed exploration of the file system, including viewing file details, timestamps, and identifying deleted files. Different features such as file analysis and metadata inspection helped reveal important information about user activity and file history. The lab also demonstrated how investigators can trace actions and recover hidden or deleted data without altering the original evidence. Overall, this lab proved how using the Autopsy platform makes it easier to analyze digital evidence and simpler to understand events that occur on a device.

Part 2: Technical summary

In this lab, we concentrated on a different area of Linux Forensics, the Autopsy Platform. Autopsy is a digital forensic browser-based platform and graphical interface to The Sleuth Kit and other tools. To ensure we have the Autopsy platform installed on our Kali Linux virtual machine, we input the command **sudo apt-get install autopsy**. Once the Autopsy platform was successfully downloaded onto my machine, I was able to dive further into the different features of the platform. The main key features Autopsy includes are as follows: image analysis, file activity timeline, image integrity, events sequencer, file analysis, metadata analysis, and report generation.

To access the Autopsy platform, we entered the following command as a root user: **sudo autopsy**.

```
(kali@kali)-[~]
$ sudo autopsy

Autopsy Forensic Browser
http://www.sleuthkit.org/autopsy/
ver 2.24

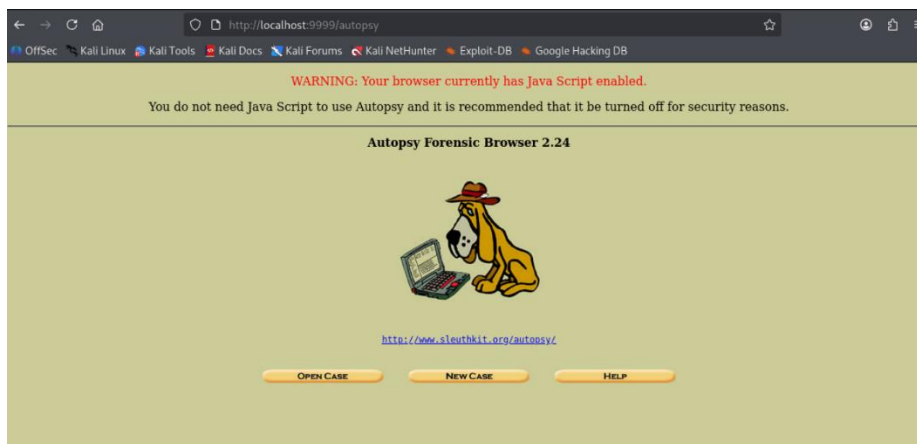
Evidence Locker: /var/lib/autopsy
Start Time: Tue Apr 7 16:48:42 2026
Remote Host: localhost
Local Port: 9999

Open an HTML browser on the remote host and paste this URL in it:

http://localhost:9999/autopsy

Keep this process running and use <ctrl-c> to exit
```

After running that command, the Autopsy Forensic Browser appeared listing the evidence locker number, as well as the start time of the platform, the remote host name, and the local port number. Without closing the Linux terminal, we need to paste the URL <http://localhost:9999/autopsy> into a new browser. The screen below is what emerges after hitting enter.



Before we click any of the “OPEN CASE” or “NEW CASE” buttons, we need to download an image to be analyzed first. I used the website provided and visited the site <https://downloads.digitalcorpora.org/corpora/scenarios/2009-m57-patents/usb/>. I double clicked on the last image named **terry-work-usb-2009-12-11.E01** and downloaded it. I saved the image to the folder **/home/kali/Downloads/**, and we are now ready to start a new case.

To start a new case, we need to go back to the Autopsy Forensic Browser website that we pasted into our virtual machine browser. I clicked on “NEW CASE” and filled in the required fields before clicking “NEW CASE” again.

CREATE A NEW CASE

1. **Case Name:** The name of this investigation. It can contain only letters, numbers, and symbols.

2. **Description:** An optional, one line description of this case.

3. **Investigator Names:** The optional names (with no spaces) of the investigators for this case.

a. SarahBenavides	b.
c.	d.
e.	f.
g.	h.
i.	j.

For the case name I used **terry-usb**, for the description I input **terry-work-usb** and I listed myself as the only investigator. After hitting the “NEW CASE” button again, another screen appears allowing us to create a new host on which results of the analysis will be saved. I chose my name and clicked “ADD HOST”. Once I added myself as the host, the configuration file **case.aut** was created. I used the **cat** command to display the contents of the file in the terminal using the command **cat /var/lib/autopsy/terry-usb/cat.aut**.

```
(kali㉿kali)-[~]
$ cat /var/lib/autopsy/terry-usb/cat.aut
# Autopsy case config file
# Case: terry-usb

created Tue Apr  7 17:35:14 2026
desc terry-work-usb
images images
data      output
log       logs
reports reports
```

This output displays the case name, when the case was created, the description we gave the case, and the empty columns for the data, output, logs, and reports.

Once we entered all the required information for adding a host such as host name, description, and time zone, we clicked the button “ADD HOST”.

1. **Host Name:** The name of the computer being investigated. It can contain only letters, numbers, and symbols.

terry-usb-host

2. **Description:** An optional one-line description or note about this computer.

Terry Host

3. **Time zone:** An optional timezone value (i.e. EST5EDT). If not given, it defaults to the local setting. A list of time zones can be found in the help files.

CT

4. **Timeskew Adjustment:** An optional value to describe how many seconds this computer's clock was out of sync. For example, if the computer was 10 seconds fast, then enter -10 to compensate.

0

5. **Path of Alert Hash Database:** An optional hash database of known bad files.

6. **Path of Ignore Hash Database:** An optional hash database of known good files.

The next window to appear shows that we are trying to add the host **terry-usb-host** to the case **terry-usb**. This window also allows us to add an image to provide the full path to the USB image we downloaded earlier. I clicked on “ADD IMAGE” and provided the full path to the image which was **/home/kali/Downloads/terry-work-usb-2009-12-11.E01**.

ADD A NEW IMAGE

1. **Location**
Enter the full path (starting with /) to the image file.
If the image is split (either raw or EnCase), then enter '*' for the extension.

me/kali/Downloads/terry-work-usb-2009-12-11.E01

2. **Type**
Please select if this image file is for a disk or a single partition.

☒ Disk ☐ Partition

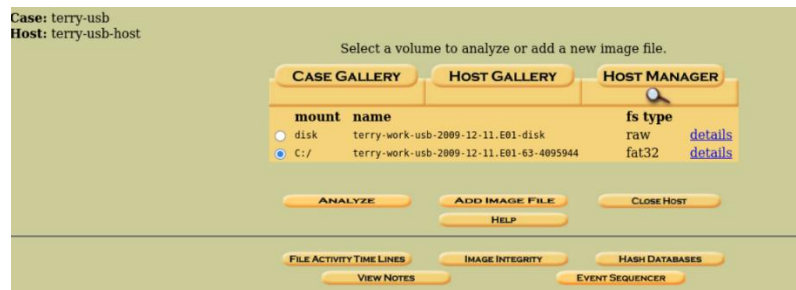
3. **Import Method**
To analyze the image file, it must be located in the evidence locker. It can be imported from its current location using a symbolic link, by copying it, or by moving it. Note that if a system failure occurs during the move, then the image could become corrupt.

☒ Symlink ☐ Copy ☐ Move

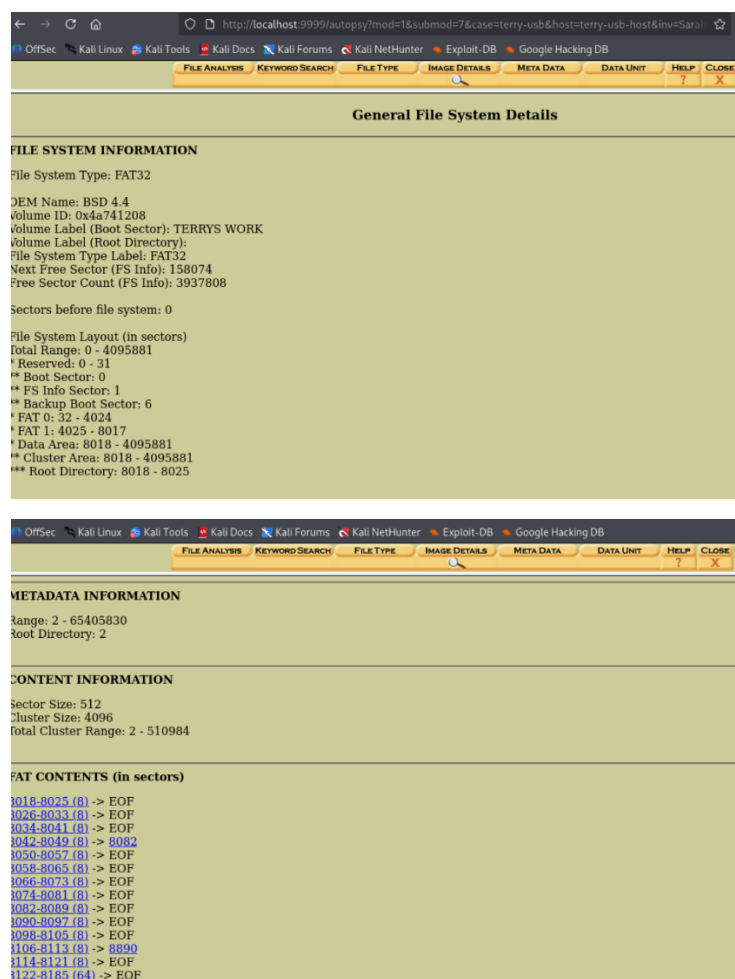
NEXT

CANCEL HELP

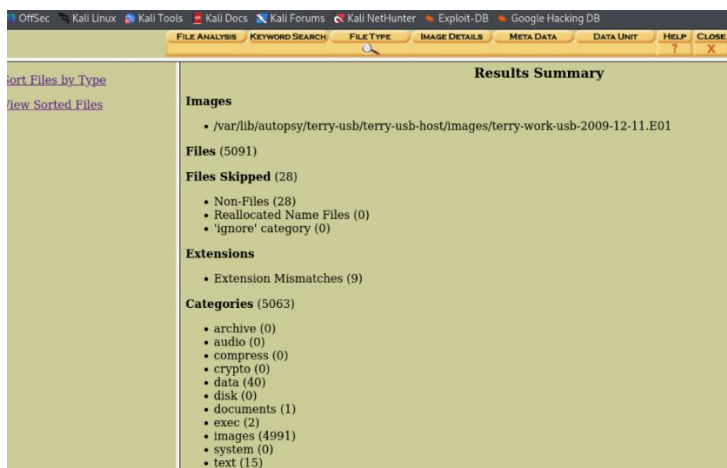
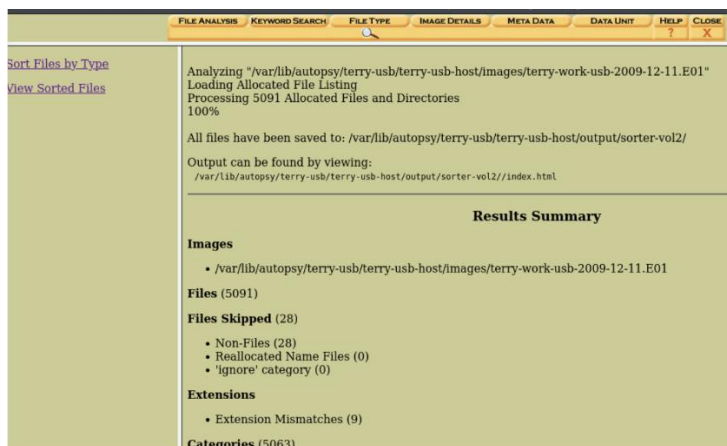
After entering the file path, I clicked “NEXT” and the image file details appeared. I ensured the mount point was C, and the file system type was fat32. I then selected the second option within the case gallery in the next window, and this is shown in the screenshot below.



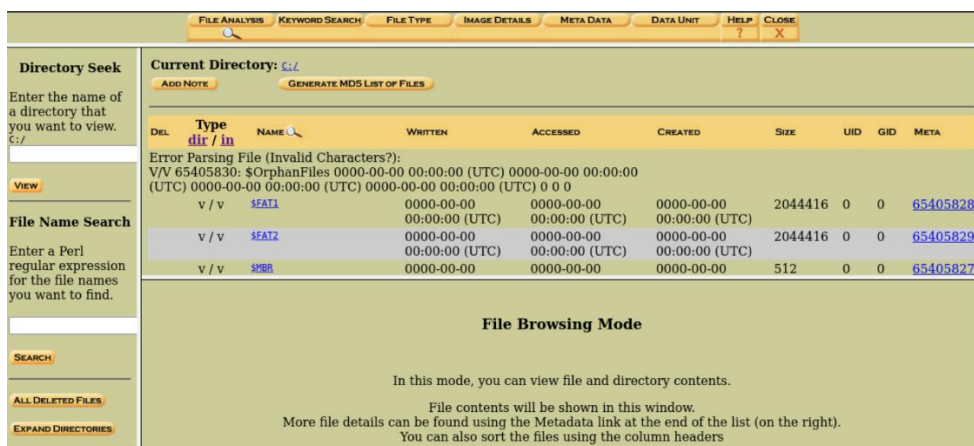
Once I selected the current option, I selected the “ANALYZE” button to begin analyzing the mounted image. Before the analysis began, we were directed to a page that made us choose which type of analysis to perform. Here I clicked on “IMAGE DETAILS” and “FILE TYPE”. When choosing image details, a list appeared displaying the file system information, the metadata information, content information and a very long list of fat contents.



When choosing file type, all files were sorted by different types within the images folder. The result summary lists the images we analyzed, the number of files analyzed, the number of files that were skipped, the extensions, and the different categories the files were sorted into.



After examining the two previous analysis types, we now will focus on the “FILE ANALYSIS” tab. This window displays about ten different columns. The columns include files that are deleted, the types and names of the files, when the files were written, accessed and created the size of the files, as well as the UID, GID, and the Meta number.



When looking at all the different time columns like written, accessed, and created, all times are zero. But when I clicked on either “dir” or “in” under TYPE, the times and dates changed to specific ones from the year 2009.

DEL	Type	NAME	WRITTEN	ACCESSED	CREATED	SIZE	UID	GID	META
	dir / in								
	Error Parsing File (Invalid Characters?): V/V 65405830: \$OrphanFiles 0000-00-00 00:00:00 (UTC) 0000-00-00 00:00:00 (UTC) 0000-00-00 00:00:00 (UTC) 0 0 0								
✓	d / d	.fseventsd/	2009-11-17 10:48:38 ()	2009-11-17 00:00:00 ()	2009-11-17 10:48:38 ()	0	0	0	10
	d / d	.Spotlight-V100/	2009-11-17 10:47:46 ()	2009-11-17 00:00:00 ()	2009-11-17 10:47:47 ()	4096	0	0	13
	d / d	.Trashes/	2009-11-17	2009-11-17	2009-11-17	4096	0	0	8

Both “dir” and “in” tabs had the same values for all the columns. After analyzing the timestamps, I clicked the button “GENERATE MD5 LIST OF FILES” to generate a list of MD5 hash values of the listed files. The hash values are below.

```
MD5 Values for files in C:/ (terry-work-usb-2009-12-11.E01-63-4095944)

d41d8cd98f00b204e9800998ecf8427e - TERRYS WORK (Volume Label Entry)
5ecad39c470178e1b0ef93e534b60fda - ._.Trashes
ce3231180e69f5a56c16459da0f1d531 - webauto.py
8ad51236530591ff2a9e84a15bd89ed1 - M57biz.jpg
7e777ea21c6763c62518c304ca5d4ce1 - . M57biz.jpg
64ee6fb48ca85ce93df69bb7941a4b54 - urlstime_machine.txt
4ee3e80d4fa851aca7ec2bb40b9b77c3 - .urlstime_machine.txt
5170ec45ba4ed423ffed57b431a1fe4f - .webauto.py
d72b25eccfe732030000ddde08335191 - urlspersona.txt
5e1783e9f289ece7d6d9c3952f38b1d1 - .urlspersona.txt
95fa678acc9369d89a4513a0b8e4b26d - urlspatents.txt
5e9fc9d520c0acf4b4calec77f80325f - .urlspatents.txt
31262fcf2a6d69d3bacb7574fc9e0445 - urlscryptography.txt
341e85a6854a69687facfdbbfd18afac - .urlscryptography.txt
02fcfbd3f7b2b021bec1d27250a832d0 - urlscopyright.txt
d4799d3877f0867c5b21f1217284aa05 - .urlscopyright.txt
35c3cd97833a1be73d974f4798bcbbf7 - patentauto.py
071a9abef36c665a4dc480098d974ff4 - .patentauto.py
934541851380f404a8efb659003c6ad7 - patentterms.txt
b02a2a4b05e80a81e38fcf8bccef1d1b - .patentterms.txt
55f9facaec8cf1d1bf1f8c04f3d287a4 - R54402.EXE
790856e8ae34f0ac3eb40e031cf7a47a - vnc-4_1_3-x86_win32.exe
```

Lastly, we examined the Meta column, there were three different numbers. I clicked each one and it brought me to a different window listing the specific file type, the MD5 of content, the SHA-1 of content, specific details of the file like directory entry, file attributes, size and name, as well as a long list of sectors.

Meta 1:

FILE ANALYSIS
KEYWORD SEARCH
FILE TYPE
IMAGE DETAILS
META DATA
DATA UNIT
HELP
CLOSE

Dir Entry Number:
65405828

VIEW

ALLOCATION LIST:

PREVIOUS
NEXT

REPORT
VIEW CONTENTS
EXPORT CONTENTS
ADD NOTE

Search for File Name

File Type:
data

MD5 of content:
b380946c6b0e090e489bef18e28d2b4e -

SHA-1 of content:
c19d8dbfa3ade5be3cb75d861364a12f35f735fb -

Details:

Directory Entry: 65405828
Allocated
File Attributes: Virtual File
Size: 2044416
Name: \$FAT1

Directory Entry Times:
Written: 0000-00-00 00:00:00 (UTC)
Accessed: 0000-00-00 00:00:00 (UTC)
Created: 0000-00-00 00:00:00 (UTC)

Sectors:
32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47
48 49 50 51 52 53 54 55


```
Autopsy Dir Entry Report

-----
GENERAL INFORMATION

Dir Entry: 65405828
Pointed to by file(s):
  C:/SFAT1
MD5 of istat output: be9180bbf50a8672c02b433f2e914a7e  -
SHA-1 of istat output: 4575823791231a354b123d5c0d89062d9ae14f52  -

Image: '/var/lib/autopsy/terry-usb/terry-usb-host/images/terry-work-usb-2009-12-11.E01'
Offset: 63 to 4095944
File System Type: fat32

Date Generated: Tue Apr  7 18:57:41 2026
Investigator: SarahBenavides

-----
META DATA INFORMATION

Directory Entry: 65405828
Allocated
File Attributes: Virtual File
Size: 2044416
Name: $FAT1

Directory Entry Times:
Written:      0000-00-00 00:00:00 (UTC)
Accessed:     0000-00-00 00:00:00 (UTC)
Created:      0000-00-00 00:00:00 (UTC)

Sectors:
32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47
48 49 50 51 52 53 54 55
56 57 58 59 60 61 62 63
64 65 66 67 68 69 70 71
72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87
88 89 90 91 92 93 94 95
```

Meta 2:

FILE ANALYSISKEYWORD SEARCHFILE TYPEIMAGE DETAILSMETA DATADATA UNITHELPCLOSE

Dir Entry Number:
65405829

VIEW

ALLOCATION LIST

◀PREVIOUSNEXT▶

REPORTVIEW CONTENTSEXPORT CONTENTSADD NOTE

[Search for File Name](#)

File Type:
data

MD5 of content:
b380946c6b6b090b489bef18e28d2b4e -

SHA-1 of content:
c19d0dbfa3ade5be3cb75d861364a12f35f735fb -

Details:
Directory Entry: 65405829
Allocated
File Attributes: Virtual File
Size: 2044416
Name: \$FAT2

Directory Entry Times:
Written: 0000-00-00 00:00:00 (UTC)
Accessed: 0000-00-00 00:00:00 (UTC)
Created: 0000-00-00 00:00:00 (UTC)

Sectors:
[4025](#) [4026](#) [4027](#) [4028](#) [4029](#) [4030](#) [4031](#) [4032](#)
[4033](#) [4034](#) [4035](#) [4036](#) [4037](#) [4038](#) [4039](#) [4040](#)
[4041](#) [4042](#) [4043](#) [4044](#) [4045](#) [4046](#) [4047](#) [4048](#)


```
Autopsy Dir Entry Report
-----
GENERAL INFORMATION
Dir Entry: 65405829
Pointed to by file(s):
C:/FAT2
MD5 of istat output: e57248b6ac7b67ff4f9d8e9f7af7b5d1 -
SHA-1 of istat output: 8875bd3ff99e35c14e49e24153150e77f7917aa7 -
Image: '/var/lib/autopsy/terry-usb/terry-usb-host/images/terry-work-usb-2009-12-11.E01'
Offset: 63 to 4095944
File System Type: fat32
Date Generated: Tue Apr 7 18:58:29 2026
Investigator: SarahBenavides
-----
META DATA INFORMATION
Directory Entry: 65405829
Allocated
File Attributes: Virtual File
Size: 2044416
Name: $FAT2
Directory Entry Times:
Written: 0000-00-00 00:00:00 (UTC)
Accessed: 0000-00-00 00:00:00 (UTC)
Created: 0000-00-00 00:00:00 (UTC)
Sectors:
4025 4026 4027 4028 4029 4030 4031 4032
4033 4034 4035 4036 4037 4038 4039 4040
4041 4042 4043 4044 4045 4046 4047 4048
4049 4050 4051 4052 4053 4054 4055 4056
4057 4058 4059 4060 4061 4062 4063 4064
4065 4066 4067 4068 4069 4070 4071 4072
4073 4074 4075 4076 4077 4078 4079 4080
4081 4082 4083 4084 4085 4086 4087 4088
```

Meta 3:

FILE ANALYSISKEYWORD SEARCHFILE TYPEIMAGE DETAILSMETA DATADATA UNITHELPCLOSE

Dir Entry Number:
65405827

VIEW

ALLOCATION LIST

Search for File Name

REPORTVIEW CONTENTSEXPORT CONTENTSADD NOTE

File Type:
DOS/MBR boot sector, code offset 0x58+2, OEM-ID "BSD 4.4", sectors/cluster 8, sectors/track 32, heads 255, sectors 4095882 (volumes > 32 MB), FAT (32 bit), sectors/FAT 3993, serial number 0x4a741208, label: "TERRYS WORK"

MD5 of content:
69de803d7609dd36123b7cabe2ae8428 -

SHA-1 of content:
40621a27984d93209a2d368506ec2399a3f72a31 -

Details:
Directory Entry: 65405827
Allocated
File Attributes: Virtual File
Size: 512
Name: \$MBR
Directory Entry Times:
Written: 0000-00-00 00:00:00 (UTC)
Accessed: 0000-00-00 00:00:00 (UTC)
Created: 0000-00-00 00:00:00 (UTC)
Sectors:
0

```
Autopsy Dir Entry Report
-----
GENERAL INFORMATION
Dir Entry: 65405827
Pointed to by file(s):
C:/MBR
MD5 of istat output: 714279cc1387c5eeb8e9230a6af090 -
SHA-1 of istat output: 8f16b2a5fba494fc48ec61a35c44013918a5247 -
Image: '/var/lib/autopsy/terry-usb/terry-usb-host/images/terry-work-usb-2009-12-11.E01'
Offset: 63 to 4095944
File System Type: fat32
Date Generated: Tue Apr 7 19:00:15 2026
Investigator: SarahBenavides
-----
META DATA INFORMATION
Directory Entry: 65405827
Allocated
File Attributes: Virtual File
Size: 512
Name: $MBR
Directory Entry Times:
Written: 0000-00-00 00:00:00 (UTC)
Accessed: 0000-00-00 00:00:00 (UTC)
Created: 0000-00-00 00:00:00 (UTC)
Sectors:
0
File Type: DOS/MBR boot sector, code offset 0x50+2, OEM-ID "BSD 4.4", sectors/cluster 8, sectors/track 32, heads 255, sectors 4095882 (volumes > 32 MB), FAT (32 bit), sectors/FAT 3993, serial number 0x4a741208, label: "TERRYS WORK"
-----
VERSION INFORMATION
```

The screenshots above are in the same order as the Meta column of the file analysis table. I also included a screenshot of a report that was generated for each Meta number.

Part 3: References

1. Lab 12: Linux Forensics (Autopsy Platform)